

Project Initialization and Planning Phase

Date	25 June 2026
Team ID	
Project Title	Human Development Index (HDI) Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project aims to develop a **Human Development Index (HDI) Prediction System** using Machine Learning techniques. The system predicts the Human Development Index of a country based on important socio-economic indicators such as life expectancy, education level, and Gross National Income (GNI). The project addresses the need for accurate and efficient HDI prediction, assisting researchers, policymakers, and educational institutions in analyzing human development trends. The solution integrates a Linear Regression machine learning model with a Flask-based web application to provide real-time HDI predictions through a user-friendly interface.

Project Overview	
Objective	To develop a Machine Learning-based Human Development Index (HDI) Prediction System that accurately predicts the HDI value using socio-economic indicators and provides an interactive web application for users to obtain predictions efficiently.
Scope	The project focuses on collecting and analyzing HDI datasets, performing data preprocessing and visualization, training a Linear Regression model, evaluating prediction accuracy, and deploying the trained model using a Flask web application for real-time HDI prediction.
Problem Statement	
Description	Human Development Index (HDI) is one of the most important indicators used to measure the overall development of a country. Calculating and analyzing HDI manually from multiple socio-economic parameters is time-consuming and requires significant data analysis. Existing methods often require technical expertise and lack an easy-to-use prediction interface.
Impact	An intelligent HDI prediction system enables faster analysis of human development, assists researchers and policymakers in decision-making, supports educational research, and provides quick predictions using historical socio-economic data.
Proposed Solution	
Approach	The proposed solution uses a Machine Learning-based Linear Regression model trained on historical Human Development Index datasets. After preprocessing and visualizing the data, the trained

	model is integrated with a Flask web application that allows users to input socio-economic indicators and instantly receive the predicted HDI value.
Key Features	• Machine Learning-based Human Development Index Prediction • Historical HDI Dataset Analysis • Data Cleaning and Preprocessing • Exploratory Data Visualization • Linear Regression Prediction Model • Flask-based Web Application • Real-time HDI Prediction • User-friendly Prediction Interface

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications	Intel Core i5 / AMD Ryzen 5 or above
Memory	RAM specifications	8 GB (Minimum), 16 GB Recommended
Storage	Disk space for data, models, and logs	20 GB Free Storage
Software		
Frameworks	Python frameworks	Flask
Libraries	Machine Learning Libraries	Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Pickle
Development Environment	IDE	VS Code / PyCharm
Data		
Data	Source, size, format	Human Development Index (HDI) Dataset (.csv) obtained from Kaggle / GitHub

